

SKMSX 2

User's Guide

A text editor that lives inside your MSX

Version 2.0 — for MSX-DOS 1 and MSX-DOS 2, MSX1 and MSX2 or later

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1 Introduction

SKMSX is a small text editor for MSX computers running MSX-DOS. It is meant for the everyday jobs that do not deserve a big program: a quick note, a batch file, a few lines of source code, a configuration file. Press one key combination and the editor opens over whatever you were doing; press ESC and you are back exactly where you were.

SKMSX 2 is a new implementation of the SKMSX 1.5 editor. The keys you know are the same. What it does:

- **Opens on demand.** Installed as a resident program, it opens with Ctrl+Shift. On ESC the interrupted program's screen, function keys and machine state are put back.
- **Keeps your text.** While resident, the text stays in memory between sessions, so you can leave the editor, run a command and come back to the same text.
- **Uses the memory your machine has.** It keeps the text in whichever is larger: a memory mapper (up to 1 MB) or video memory (about 5 KB on an MSX1, 36 to 110 KB on an MSX2). MSX-DOS 1 needs no mapper support program for this.
- **Reads and writes plain files.** Files are saved byte for byte, with no end-of-file marker and no padding added. Lines ending in CR LF, LF or a lone CR are all understood.
- **Scrolls only the line you are on.** When a line is wider than the 40-column screen, only that line moves sideways; the other lines stay where they are.

2 Requirements

- An MSX1, MSX2 or later computer with a disk drive and **MSX-DOS 1 or MSX-DOS 2**.
- The program file `SKMSX.COM`, about 9.6 KB.
- **Text memory**, one of: a memory mapper (see section 10), or video memory of at least 16 KB.
- For the resident editor, free memory at the top of the 64 KB address space ("page 3"): about **1.7 KB** on an MSX2 or with a memory mapper, and about **5.6 KB** on an MSX1 without a mapper. If there is not enough, SKMSX tells you (see section 12).

SKMSX always draws its own 40-column text screen. On an MSX2 you can start it from an 80-column screen; that screen is restored when you leave.

MSX-DOS 1 and a memory mapper. MSX-DOS 1 has no memory services of its own, so SKMSX looks for a mapper (for example a 512 KB or 4 MB RAM cartridge) by itself and uses every segment that MSX-DOS 1 does not use. You do not need a mapper support program. If one is installed (such as Konamiman's MSR) or you run MSX-DOS 2, SKMSX asks it for its segments instead.

3 Starting SKMSX

3.1 Quick editing, once

Type the program name at the DOS prompt:

```
A>skmsx
```

SKMSX prints a banner telling which memory it uses and how much text fits, then opens the editor:

```
SKMSX 2 development - Memory mapper, usable bytes: 1029120
ESC returns to DOS.
```

When you press ESC, the editor closes and **the text is discarded** (the memory is given back). Save with F2 or F3 first. This mode is the one to use for a single job.

3.2 Resident, opened with Ctrl+Shift

To keep SKMSX in memory, install it with the /R switch:

```
A>skmsx /R
SKMSX 2 resident - Memory mapper, usable bytes: 1029120
Ctrl+Shift opens; ESC closes; /U removes.
```

From now on, hold **Ctrl** and **Shift** together to open the editor, and press **ESC** to return. Your text stays in memory between sessions. Installing lowers the top of memory a little, so programs started afterwards see slightly less free RAM.

3.3 Switches

Command	What it does
SKMSX	Open the editor once. The text is lost when you press ESC.
SKMSX /F name	Open once with the file name already loaded.
SKMSX /R	Install the resident editor. It opens with Ctrl+Shift.
SKMSX /R /F name	Install resident. The file name is loaded the first time the editor opens.
SKMSX /U	Remove the resident editor and free its memory. Only possible while SKMSX is the most recently installed resident program.

- Switches are not case sensitive and can come in any order: SKMSX /F NOTES.TXT /R and SKMSX /R /F NOTES.TXT are the same.
- /F takes the next word as the file name, up to a space or the next /. A drive letter is allowed (SKMSX /F B:LETTER.TXT). File names are plain 8.3 names. A name without a drive letter means the drive that is current when the editor opens: for /R /F, when you first press Ctrl+Shift, so write the drive if the file is not on the current one.

- If the file does not exist, the editor opens with an empty text and takes that name, so F2 creates it (the status line says "New file"). If the name is not valid, the status line says "Bad 8.3 name".
- /U wins over every other switch. Unknown switches are ignored.
- Without /R, SKMSX never stays in memory.

4 The editor screen

The screen has 40 columns and 24 rows:

Rows	Content
1–22	Your text. The cursor is the # character drawn over the text.
23	A ruler: marks every tenth column, - the columns between.
24	The status line.

4.1 The status line

NOTES.TXT I M 405 Saved

Columns	Meaning
1–14	Name of the file being edited (Untitled until it has a name).
16	I = insert mode, O = overwrite mode (Insert key).
18	V = the text is in video memory, M = in a memory mapper.
20–26	Size of the text in bytes.
28–40	The last message ("Loaded", "Saved", "Not found" ...). See section 11.

4.2 How text is shown

- A TAB character is shown as spaces up to the next multiple of 8 columns.
- Other control characters are shown as a dot (.). They are still in the text and are saved unchanged.
- **Horizontal scrolling follows the cursor line only.** When the cursor goes beyond column 40, that line scrolls sideways to keep the cursor on screen. Every other line is shown from its first column and does not move. When you move to another line, it is shown with its own position: a line that fits on the screen always starts at column 1, and a long line puts the cursor near the middle.

5 Moving around

Key	Moves the cursor
Left / Right	One character. At the end of a line, Right goes to the start of the next.
Up / Down	One line up or down, keeping the column (see below).
Shift + Left / Right	To the start of the previous / next word.
Shift + Up / Down	Scrolls the text one line; the cursor moves with it.
Ctrl + Left / Right	To the start / end of the line.
Ctrl + Up / Down	18 lines up / down (a "page").
Home (also Ctrl + Home)	To the start of the text.
Shift + Home	To the end of the text.

Words are separated by spaces. A line break counts as the end of a word.

Keeping the column. When you press Up or Down repeatedly, SKMSX remembers the column you started from. Passing through a short line puts the cursor at its end, but the next long line brings it back to the original column. Any other key forgets the remembered column. If the remembered column falls in the middle of a TAB, the cursor stops just before the TAB.

On the first line Up does nothing, and on the last line Down does nothing.

6 Editing text

- **Typing** inserts characters at the cursor.
- **Insert** switches between insert and overwrite mode (**I** / **O** on the status line). In overwrite mode a typed character replaces the one under the cursor, except at the end of a line or of the text, where it is simply added.
- **Enter** inserts a line break as two bytes, CR and LF.
- **Tab** inserts a TAB character.

Key	Deletes
Backspace	The character before the cursor.
Delete	The character under the cursor.
Shift + Backspace	Back to the start of the word (spaces before the cursor and the word itself).
Shift + Delete	Forward to the end of the next word (any spaces at the cursor, then the word). The spaces after the word stay.
Ctrl + Backspace	Everything between the start of the line and the cursor.
Ctrl + Delete	Everything between the cursor and the end of the line.

Backspace and Delete remove a CR LF pair as one character, joining the two lines. Word deletion at the very start or end of a line removes the line break and joins the lines. Deleting to the start or end of a line never goes past the line break.

When memory is full, SKMSX says "Buffer full" and refuses the insertion: the text is never damaged.

7 Files

Key	Action
F1	Load a file, or start a new one. Asks Load/new: .
F2	Save under the current name. If the text has no name yet, asks for one like F3.
F3	Save under another name. Asks Save as: .

7.1 Loading with F1

Where the cursor is decides what F1 does:

Cursor	Result
At the start of the text (Home)	The file replaces the text in memory. The old text is discarded without asking. If the file does not exist, the text is cleared and takes the new name ("New file"): this is how you start a new file.
At the end of the text	The file is added after the existing text.
Anywhere else	Nothing is loaded; the status line says "Use HOME/EOF".

The file must fit in the free memory. The text already in memory counts until it is replaced, so to load a very large file over a large text, save and clear the old text first. If it does not fit, the status line says "Buffer full" and the text is left as it was.

7.2 Saving with F2 and F3

- File names are plain **8.3 names**, with an optional drive letter from A: to H: in front (B:REPORT.TXT). Paths with directories are not supported. Names are saved in upper case. These characters are not allowed: " * + , / : ; < = > ? [\] |.
- The text is written **exactly as it is in memory**: SKMSX adds no end-of-file marker and no padding.
- The save is safe. SKMSX writes the new contents to a temporary file SK2SAVE. \$\$\$, renames the existing file to SK2BAK. \$\$\$, renames the temporary file to the real name, and only then deletes the backup. If anything fails, the original file is still there.

If you see "**SK2 temp used**", a SK2SAVE. \$\$\$ or SK2BAK. \$\$\$ file from an earlier interrupted save is still on the disk. Delete or rename it and save again. If you see "**Keep SK2BAK!**", the last step

failed and could not be undone: both files have been kept. Rename SK2BAK . \$\$\$ back to the original name to recover your old file.

8 Search and replace

Key	Action
F4	Find. Asks Find:, then moves the cursor to the start of the next match.
F9	Repeat the last find, starting after the cursor.
F5	Find and replace. Asks Find: and With:, then replaces the next match.
F10	Repeat the last find and replace.

- The search runs **forward** from the cursor to the end of the text. F4 and F5 also match a word that starts exactly at the cursor; F9 starts one character later. It is **case sensitive** and takes the characters exactly as typed (no wildcards). The search text and the replacement can be up to 31 characters.
- The status line says "Found" or "Not found". When nothing is found, the cursor does not move.
- After a replacement, the cursor is placed just after the new text, so F10 continues with the next match.
- An empty with: deletes the matches. If the replacement does not fit in memory, the status line says "Buffer full" and nothing is changed.
- Press ESC at any prompt to cancel it ("Cancelled"). This only cancels the prompt: the editor stays open.

9 Blocks

A block is a piece of text that you can copy to another place. SKMSX has one block at a time.

Key	Action
F6	Mark the start of the block at the cursor ("Block start").
F7	Mark the end of the block at the cursor ("Block marked"). It must be after the start.
F8	Copy the block to the cursor position.

- Move the cursor to where the copy should go and press F8. The status line says "Mark a block" when there is no complete block, and "Inside block" when the cursor is inside the block (the text cannot be copied into itself).
- The marks are forgotten as soon as you change the text (typing, deleting, Enter, replace), and after each copy. To copy again, mark the block again.

- A copy needs free memory for the whole block; otherwise "Buffer full" is shown and nothing is copied.

10 Memory and capacity

SKMSX chooses where to keep the text when it starts, and tells you in the banner:

Storage	Status	Capacity
Memory mapper	M	Up to 1 MB (64 segments of 16 KB). Used whenever it is larger than the video memory: with MSX-DOS 2 or a mapper support program through them, with plain MSX-DOS 1 found by SKMSX itself.
Video memory, MSX2 with 128 KB	V	About 100 KB resident, 110 KB in quick mode.
Video memory, MSX2 with 64 KB	V	About 36 KB resident, 45 KB in quick mode.
Video memory, MSX1 (16 KB)	V	About 5 KB.

- SKMSX uses the **larger** of the best memory mapper and the video memory. A mapper needs at least two free segments (three for the resident editor).
- Part of the memory is kept for SKMSX itself: a screen backup and the state of DOS and the BIOS while the editor is open (3 KB with MSX-DOS 1, 16 KB with MSX-DOS 2).
- Mapper segments are given back when the editor ends (quick mode) or is removed with /U.
- The video memory figures above are for MSX-DOS 1. With MSX-DOS 2 the reserved part is 13 KB larger, so take 13 KB off.
- The size shown in the banner is the exact number of bytes you can type.

10.1 Where the resident editor program is kept

Only a small core stays in ordinary RAM (about 1.7 KB); the editor program itself is kept elsewhere and brought in when you press Ctrl+Shift:

Machine	Editor program kept in	Ordinary RAM used
Memory mapper chosen	one mapper segment, switched in for each session	about 1.7 KB
MSX2, video memory chosen	video memory; copied into RAM at 4000h for each session, and the program RAM it covers is saved and put back afterwards (about 9 KB of video memory is used for this)	about 1.7 KB
MSX1 without a mapper	ordinary RAM, as one block	about 5.6 KB

Another program using the same mapper memory. With MSX-DOS 2 or a mapper support program, the operating system hands out mapper segments, so programs do not collide. Plain MSX-DOS 1 cannot do that: SKMSX takes every segment that MSX-DOS 1 itself does not use. Do not combine it with another program that keeps data in the same mapper (a RAM disk, for example), or install a mapper support program first.

Video memory storage and graphics programs. In video memory mode the text lives in the video RAM that a graphics program may be using (on an MSX1, 2020h–3FFFh; on an MSX2, 4000h and up). That part of the picture or data is not restored when you leave the editor. DOS and BASIC text screens are not affected. With a mapper this problem does not exist.

11 Messages

11.1 Status line

Message	Meaning
Loaded	The file was read.
Saved	The file was written.
New file	F1 or /F named a file that does not exist; the text starts empty with that name.
Found / Not found	Result of a search or replacement.
Cancelled	A prompt was cancelled with ESC.
Block start / Block marked	F6 or F7 recorded a position.
Mark a block	F7 or F8 was used without a valid block.
Inside block	F8 refused: the cursor is inside the block.
Use HOME/EOF	F1 can only be used with the cursor at the start or at the end of the text.
Buffer full	Not enough free memory. Nothing was changed.
Bad 8.3 name	The file name is not valid.
Disk error	DOS reported an error (file not found, disk full, write protected ...).
SK2 temp used	Leftover SK2SAVE .\$\$\$ or SK2BAK .\$\$\$ found; see section 7.2.
Keep SK2BAK!	Saving failed half-way; both files were kept; see section 7.2.

11.2 Messages when starting SKMSX

Message	Meaning and remedy
SKMSX 2 is already resident.	The resident editor is already installed.
SKMSX 2 hook not found.	/U was used but SKMSX is not installed.
Remove later residents first.	Another resident program was installed after SKMSX. Remove that

Message	Meaning and remedy
	one first, then /U.
Insufficient page-3 RAM for resident editor.	There is not enough free memory at the top of RAM; see section 12.
SKMSX 2 removed.	/U worked.

12 Problems and limitations

12.1 "Insufficient page-3 RAM for resident editor."

The resident editor needs about 1.7 KB of free memory at the top of RAM on an MSX2 or with a memory mapper, and about 5.6 KB on an MSX1 without a mapper. Other resident programs (network or SD-card interfaces, RAM disks, ...) also take memory there, so this message can only appear on an MSX1 without a mapper, or when page 3 is almost full. Try, in this order:

1. **Add a memory mapper** (a RAM cartridge): SKMSX finds it by itself and then needs only about 1.7 KB.
2. Do not load the other resident programs you do not need.
3. Use the quick mode (SKMSX without /R), which needs no resident memory.

12.2 Other limits

- One text at a time; there are no split screens or windows.
- File names are plain 8.3 names with an optional drive letter; directories are not supported.
- The editor screen is always 40 columns wide. Lines wider than that scroll sideways (section 4.2).
- In quick mode the text disappears when you press ESC. Save first.
- /U only works if SKMSX is the most recently installed resident program.
- Video memory storage can overwrite what a graphics program keeps in video RAM (section 10).
- On plain MSX-DOS 1, SKMSX cannot know which mapper segments other programs use (section 10.1).

Appendix A Compiling SKMSX

SKMSX is written in Z80 assembly language. The whole program is built with one command and produces a single SKMSX.COM.

A.1 What you need

- **SjASMPPlus**, a Z80 cross-assembler (<https://github.com/z00m128/sjasmplus>). SKMSX is built with version 1.24.0. The build uses SjASMPPlus' relocation directives, so use a recent version.
- Nothing else. The source files are in SKMSX_v2.0 (skmsx.asm and the src folder).

A.2 Building on Windows with make.bat

Open a command prompt in the SKMSX_v2.0 folder and run:

```
make          build SKMSX.COM
make clean    delete SKMSX.COM and skmsx.sym
```

make.bat looks for the assembler in this order: the environment variable SJASMPPLUS (full path of sjasmplus.exe), the folder C:\Users\roniv\Dev\bin, then the PATH. To use a copy somewhere else:

```
set SJASMPPLUS=D:\tools\sjasmplus\sjasmplus.exe
make
```

On success it prints the size of the file:

```
Errors: 0, warnings: 0, compiled: 3790 lines, work time: 0.031 seconds
Built SKMSX.COM, 9643 bytes
```

On an error it prints the assembler's messages followed by BUILD FAILED and ends with exit code 1, so it can be used from other scripts.

If Windows answers "'make.bat' is not recognized" even though you are in the right folder, your system has the setting NoDefaultCurrentDirectoryInExePath turned on, which stops the command prompt from looking in the current folder. Type .\make instead of make.

A.3 Building on other systems

Run the assembler directly, from the SKMSX_v2.0 folder:

```
sjasmplus --nologo --sym=skmsx.sym skmsx.asm
```

A.4 What is produced

File	Description
SKMSX.COM	The program, an MSX-DOS .COM file (about 9.6 KB). Copy it to your MSX disk.
skmsx.sym	Symbol list (addresses of every label). Only useful for debugging in an emulator or monitor; not needed to run SKMSX.

A.5 Source files

File	Contents
skmsx.asm	Top level: output file, the memory layout of the resident program and its regions, the limit on mapper segments (MAX_SEGMENTS).
src/offsets.asm	24-bit position arithmetic.
src/buffer.asm	The text buffer (a gap buffer).
src/navigation.asm	Cursor movement and deletion by character, word and line.
src/commands.asm	Key handling, function keys, prompts, search, replace, blocks.
src/view.asm	Drawing the screen, horizontal scrolling, ruler and status line.
src/format.asm	Number formatting.
src/files.asm	Loading and saving, file name checking, /F loading.
src/platform.asm	Switching the screen mode, saving and restoring the screen, font and function keys; keyboard.
src/session.asm	One editor session from start to ESC.
src/core.asm	Services that always stay in ordinary RAM: DOS calls, video addressing, file buffers.
src/resident.asm	The Ctrl+Shift hook and the saving and restoring of the interrupted program.
src/storage.asm, src/memory.asm, src/memory_init.asm	Choosing and using the text memory (video memory or memory mapper).
src/dos2.asm, src/dos2_context.asm	MSX-DOS 2 calls, memory mapper paging and the DOS 2 state saved while the editor is open.
src/install.asm	The installer: switches, installing, relocating, removing.
src/foreground.asm	Quick mode (without /R).

A.6 How the build works

- One assembly produces one SKMSX.COM. The part that stays in memory is assembled once; at install time the installer moves it to the top of RAM and adjusts every address in it. The list of addresses to adjust is generated by SjASMPPlus itself, so nothing has to be maintained by hand.
- The editor proper is placed in a mapper segment or, on an MSX2, in video memory, and only a small core stays in ordinary RAM. On an MSX1 without a mapper everything stays in ordinary RAM as one block.
- The build stops with an error if the editor part grows beyond 16 KB or the program no longer fits below address 8000h.

A.7 Trying the result

Copy SKMSX.COM to a disk that your MSX or emulator can read, for example the folder you give to

openMSX with -diska, and run it:

```
A>skmsx
A>skmsx /F README.TXT
A>skmsx /R
```

Appendix B Quick reference

Key	Action
Ctrl + Shift	Open the resident editor.
ESC	Leave the editor (cancels a prompt first).
Arrows	Move one character or line.
Shift + Left / Right	Previous / next word.
Shift + Up / Down	Scroll one line.
Ctrl + Left / Right	Start / end of line.
Ctrl + Up / Down	18 lines up / down.
Home	Start of text.
Shift + Home	End of text.
Insert	Insert / overwrite.
Backspace / Delete	Delete a character.
Shift + Backspace / Delete	Delete a word.
Ctrl + Backspace / Delete	Delete to the start / end of the line.
F1	Load, new, or append a file.
F2	Save.
F3	Save as.
F4 / F9	Find / find again.
F5 / F10	Find and replace / replace again.
F6 / F7	Mark block start / end.
F8	Copy the block to the cursor.

Command line	Effect
SKMSX	Edit once.
SKMSX /F name	Edit once, loading name.
SKMSX /R	Install resident.
SKMSX /R /F name	Install resident, loading name on first use.
SKMSX /U	Remove the resident editor.